

# Bridgeway Capital EDGAR: Insider Trading

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# Outline

- 1 Introduction
- 2 Data
- 3 Universe and benchmark
- 4 Multi-insider purchases
- 5 Small-cap purchases and seller silence
- 6 Conviction composite

## Question

Do public insider filings (SEC Forms 3/4/5) predict subsequent returns of U.S. biotech stocks?

- Insiders know their pipeline; the filing becomes public within two business days
- Literature: purchases informative, sales much less; stronger in small firms (Lakonishok–Lee 2001; Cohen–Malloy–Pomorski 2012)
- Our constraint: only what a trader could know on the filing date, and keep the firms that later delisted

# Project specification

- **Universe:** U.S. biotech common stock, point-in-time, delisted firms included
- **Timing:** information date = SEC filing date; entry at the next session's close
- **Horizons:** 1, 2, 3, 5, 10, 21, 42, 63, 126 sessions
- **Benchmark:** matched XBI total return (small-cap purchases, seller silence, conviction); equal-weight biotech (multi-insider purchases)
- **Deliverables:** reusable data pipeline, then signals tested baseline-first, with results reported whether positive or null

Pipeline: Scrape → Universe → Prices → Alpha

# Data: SEC Form 3/4/5/144 parser

One engine per form, all on one shared format-aware parser

| Engine        | Form                                | Rows are...  |
|---------------|-------------------------------------|--------------|
| Form3Engine   | 3: initial ownership snapshot       | holdings     |
| Form4Engine   | 4: changes in ownership (workhorse) | transactions |
| Form5Engine   | 5: annual exempt or late trades     | transactions |
| Form144Engine | 144: notice of proposed sale        | sale notices |

- Two sources: EDGAR filing by filing back to 2003, SEC quarterly data sets from 2006
- Information date = SEC filing date: no trade is used before it was public
- Every row links back to its original filing (accession, filing date, 10b5-1 flag)
- Form 144 parser ready; Form 144 filings not yet collected

# Data: format-aware parsing

Twenty years of filings in four formats, one output table

| Era      | Format           | Status        |
|----------|------------------|---------------|
| 2004 →   | XML table layout | ok            |
| 2003     | XML flat X0101   | ok_legacy_xml |
| 2003     | HTML render      | partial_html  |
| pre-2003 | ASCII text       | partial_txt   |
| any      | unreadable       | error (stub)  |

- Each filing's format is detected, then sent to the matching reader
- Older formats are only partly readable: their rows are marked partial, unreadable filings are kept as stubs

# Data: CRSP prices

Daily CRSP data for the 953 universe stocks, 2003–2025, through WRDS

| What we collected                   | Used for                            |
|-------------------------------------|-------------------------------------|
| Daily total return, incl. delisting | excess returns at every horizon     |
| Daily low and high                  | checking reported trade prices      |
| Market cap (lagged)                 | small-cap cutoff                    |
| Split factors                       | ownership continuity across splits  |
| XBI total return                    | benchmark over the same sessions    |
| Trading calendar                    | entry at the close after the filing |

- 2.05 million stock-days, including firms later delisted or acquired
- Each filing is matched to the right stock on its date (SEC CIK → Compustat → CRSP)
- Only data known on the filing date: market caps are lagged, size cutoffs use prior years only

# Data: quality checks

Every reported trade is checked against market data and flagged if it fails

| Check, all universe trades 2006–2025  | Purchases             | Sales |
|---------------------------------------|-----------------------|-------|
| Price within the day's CRSP low–high  | 90.5%                 | 98.5% |
| Price within the week's CRSP low–high | 94.5%                 | 99.4% |
| Holdings add up across filings        | 68% of checkable rows |       |

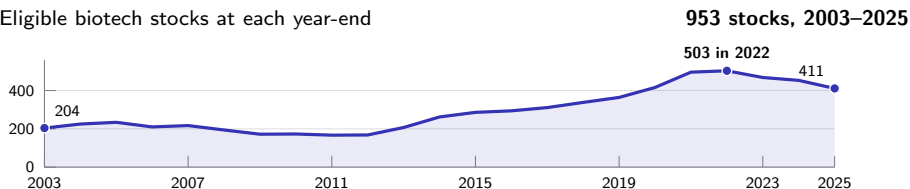
- Flagged trades are kept and marked, never deleted or corrected
- Prices outside the range are mostly purchases about 10% below the low, consistent with insiders buying in discounted share offerings
- Holdings that do not add up come from missing links in the filing chain (trusts, option exercises, Form 5 gifts), not from wrong trades

# Data: SEC, FDA and ClinicalTrials.gov data store

- Local, as-filed DuckDB store shared by all ingestion paths
- Sources: SEC bulk and filing indexes, openFDA submissions, ClinicalTrials.gov study records, SEC 8-K event candidates
- Archived raw bytes with hashes; idempotent, restartable, incremental day/week/month updates
- Purpose: catalyst conditioning (FDA decisions, trial milestones) using only publicly time-stamped information
- Status: built and tested; not yet used in the signal results

# Universe: biotech only, per the project scope

Eligible biotech stocks at each year-end



| Project requirement               | Rule used                                      |
|-----------------------------------|------------------------------------------------|
| Biotech stocks only               | → GICS Biotechnology; SIC 2836 if GICS missing |
| Major US exchange at a given date | → NYSE, NASDAQ, AMEX; dated membership         |
| CRSP share codes 10 and 11        | → US common shares: CRSP codes 10 and 11       |
| Stocks which no longer exist      | → Kept while listed, even if later delisted    |
| Full biotech history              | → CRSP and Compustat via WRDS, 2003–2025       |

The conviction composite and the 20-year study also use SIC 2834/2836. Code: [shared/universe/universe.py](#)

# Benchmark: four options in the project scope

## Benchmark options listed in the project scope

| Option                                                 | Used by                                                            |
|--------------------------------------------------------|--------------------------------------------------------------------|
| 1 Equal-weight return of all biotech stocks (own list) | <b>Multi-insider purchases</b> (company excluded)                  |
| 2 IBB (iShares)                                        | —                                                                  |
| 3 XBI (SPDR)                                           | Small-cap purchases, seller silence, conviction (matched sessions) |
| 4 Other biotech industry returns                       | —                                                                  |

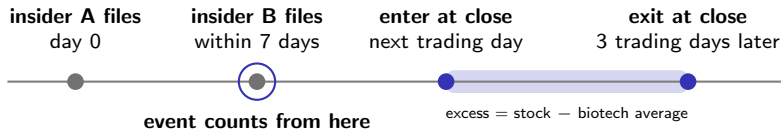
- Multi-insider purchases hold on both: +0.90 pp vs equal-weight ( $p = .031$ ); +0.96 pp vs XBI ( $p = .024$ , robustness row)
- Conservative: daily-rebalanced equal weighting carries bid-ask bounce, which lifts the benchmark and understates excess returns

Project scope, 7 Sep 2026; [signals/multi\\_insider\\_purchases/results.json](#)

# Multi-Insider Buys: two insiders within a week

One event

7-day window · 2+ buyers · 3 trading days



## Who and what qualifies

- Form 4 code P purchases only
- Individual officers and directors; no firms, funds
- Count only after the second report is filed
- Equal-weight biotech average, company excluded
- **1,683 events after removing overlaps**

## Why this rule

- Alldredge & Blank 2019: clustered purchases informative
- Lakonishok & Lee 2001: purchases, not sales
- Project scope: trades by multiple people more informative

[signals/multi\\_insider\\_purchases/README.md](#)

# Multi-Insider Buys: chosen on train, then tested

Split by filing date

each return window ends inside its period

| Train      | Validation | Test       |
|------------|------------|------------|
| 2009–2016  | 2017–2019  | 2020–2025  |
| 665 events | 296 events | 722 events |

Top 3 from 2009–16, checked on 2017–19

keep  $t > 0$ , one-sided  $p < .10$

| Window        | Buyers   | Horizon       | Estimator       | Train pp     | Valid. pp    | Valid. $t$  |                 |
|---------------|----------|---------------|-----------------|--------------|--------------|-------------|-----------------|
| 7 days        | 2        | 1 day         | mean            | +0.81        | −0.10        | −0.44       | failed          |
| <b>7 days</b> | <b>2</b> | <b>3 days</b> | <b>mean</b>     | <b>+1.16</b> | <b>+1.29</b> | <b>1.96</b> | <b>selected</b> |
| 14 days       | 2        | 5 days        | winsorized mean | +1.43        | +1.45        | 1.71        | lower $t$       |

- Grid: 7 or 14 days  $\times$  2 or 3 buyers  $\times$  7 horizons  $\times$  2 estimators = 56
- Test 2020–2025: pass = estimate  $> 0$  and two-sided  $p < .05$
- Standard error: largest of four estimates (two-way, issuer, quarter, HC1)

signals/multi\_insider\_purchases/README.md, train\_ledger.csv

# Multi-Insider Buys: +0.90 pp over 3 days, 2020–2025

**Baseline first:** one buyer +0.41 pp ( $n = 1,968$ ) vs two or more +0.90 pp ( $n = 722$ ); gap +0.49 pp,  $p = .29$ , not significant

Return above biotech average, 95% intervals, pp

## Train

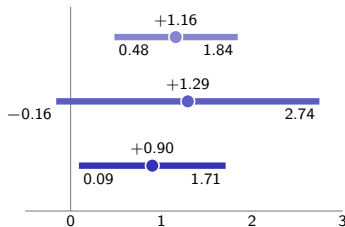
665 events

## Validation

296 events

## Test

722 events



**+0.90 pp**

**2.30**

**0.031**

2020–2025 average

t-statistic

p, two-sided

**722**

events from 251 companies, 24 quarters

Average in 2020–22 and 2023–25, pp

2020–2022 +0.48 pp

2023–2025 +1.38 pp

Drop any one company:  
+0.78 to +1.01 pp

**Next:** agree on a start date; test on new filings

Before trading costs. [signals/multi\\_insider\\_purchases/RESULTS.md](#)

# Small Caps and Seller Silence: Method

## Rules shared by both signals

- Events: Form 4 purchases (code P, including offering participation) and sales (code S)
- Information date = when the trade, or the silence, becomes public; entry at the next close; return in excess of XBI over the same sessions
- Sample 2009–2025: 2006–2008 only builds history (first size cutoffs, insiders' past selling)
- Standard errors clustered by company and by month

# Small Caps: insider purchases predict a short drift

**Signal:** an insider purchase in a company in the bottom third by market cap (about \$100M; cutoffs from prior years only), held 5 sessions

| 5 sessions vs XBI | <i>N</i> | Mean            | <i>t</i>    |
|-------------------|----------|-----------------|-------------|
| 2009–2025         | 3,524    | <b>+1.46 pp</b> | <b>4.02</b> |
| 2009–2019         | 1,729    | +1.91 pp        | 3.79        |
| 2020–2025         | 1,795    | +1.03 pp        | 2.07        |

- Holds after controlling for the prior week's return, and without flagged trades (+1.12 pp,  $t = 3.07$ )
- Found and frozen on 2009–2025, then rebuilt on the final universe; 2026 is kept untouched as the true out-of-sample test

# Small Caps: smaller firms, shorter horizons

All purchases 2009–2025, excess return vs XBI; descriptive, full sample

## By firm size (5 sessions)

| Market cap   | Mean            | <i>t</i>   |
|--------------|-----------------|------------|
| Bottom third | <b>+1.46 pp</b> | <b>4.0</b> |
| Middle third | +0.82 pp        | 3.0        |
| Top third    | +0.13 pp        | 0.8        |

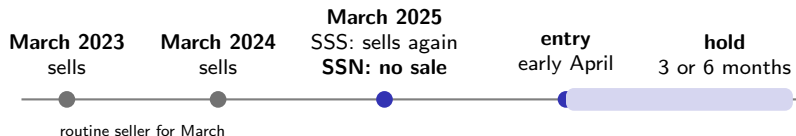
## By horizon (bottom third)

| Sessions | Mean            | <i>t</i>   |
|----------|-----------------|------------|
| 1        | +0.54 pp        | 4.1        |
| 5        | <b>+1.46 pp</b> | <b>4.0</b> |
| 10       | +2.26 pp        | 2.1        |
| 21       | +1.52 pp        | 1.2        |
| 63       | +0.59 pp        | 0.3        |

- The drift shrinks with firm size and is absent in large caps
- It peaks around two weeks, then fades: not significant after a month

# Seller Silence: why it could work

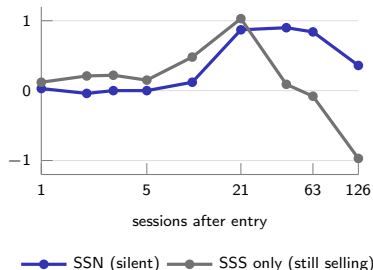
**Signal (Hong & Li, 2019):** a routine seller who stops selling



- **Firm-months:** SSN if at least one routine seller is silent; SSS only if all routine sellers sold (the comparison group)
- **Routine sales say little:** diversification, taxes, pre-planned selling
- **Stopping can say a lot:** an insider expecting good news avoids selling too cheap, and not selling is legal and invisible
- **Biotech:** binary catalysts (trial readouts, FDA decisions) are seen months ahead; a 3–6 month hold matches that timeline

# Seller Silence: results

Excess return vs XBI, 2009–2025, pp



| Firm-months       | 3 months     | 6 months     |
|-------------------|--------------|--------------|
| SSN (2,805)       | +0.84        | +0.36        |
| SSS only (1,460)  | −0.08        | −0.97        |
| <b>Difference</b> | <b>+0.91</b> | <b>+1.33</b> |
| <i>t</i>          | 0.8          | 0.7          |

Promising, but too little data

Right direction, not significant: at this effect size, about 6 times more observations would be needed for  $t = 2$ . Next: retest on a wider universe.

# Conviction: how the score is built

Score each open-market insider buy 0–4, +1 for each:

| Ingredient | Condition                                         |
|------------|---------------------------------------------------|
| Small cap  | below the rolling median market cap of prior buys |
| Big stake  | adds >10% to the insider's own holding            |
| Breadth    | clustered buying or $\geq 2$ joint owners         |
| Seniority  | officer or C-suite                                |

- Equal weights; high conviction = score  $\geq 3$
- Prediction window: 3 months (1-quarter hold, 63 sessions)
- Returns in excess of XBI, bias-free CRSP prices
- Alpha  $\approx$  strategy return  $- \beta \times$  benchmark return

# Conviction: SIC 2834/2836 walk-forward (759 firms)

Rolling window: each month the size cutoff is refit on earlier buys only, so all of 2007–2025 is out-of-sample.

| Book                 | CAGR   | Sharpe | Alpha vs XBI  | $t$ ( $p$ )        | Beta  |
|----------------------|--------|--------|---------------|--------------------|-------|
| L/S hi-conv – sells  | +10.6% | 0.61   | <b>+12.8%</b> | <b>2.80</b> (.005) | −0.02 |
| High-conviction long | +35.9% | 0.85   | +31.8%        | 2.95 (.003)        | 0.87  |
| XBI                  | +11.9% | 0.55   | —             | —                  | 1.00  |

- The L/S matches XBI's return with zero market exposure: its edge is alpha, not beta
- Event level (per trade): high conviction +3.1%/qtr vs XBI,  $p \approx .077$
- **Significant as a portfolio, marginal per trade**

walkforward\_conviction.py; gross of costs

# Conviction: how the long/short is built

- **Long leg:** buy high-conviction stocks the day after the insider-buy filing, hold  $\sim 1$  quarter, equal weight
- **Short leg:** short every stock with an insider sale the day after the filing, hold  $\sim 1$  quarter, equal weight
- Each day, positions are recomputed from the filings of the last quarter
- The short leg has no edge of its own; it is there to make the book beta-neutral

## Result

A low-beta, volatile strategy that is uncorrelated with biotech

# Conviction: SIC and GICS intersection (495 firms)

GICS 35201010, point-in-time, pharma excluded; 33,449 insider events  
2006–2025

| Book                 | CAGR   | Sharpe | Max DD | Alpha         | $t$ ( $p$ ) | Beta  |
|----------------------|--------|--------|--------|---------------|-------------|-------|
| L/S hi-conv – sells  | +7.0%  | 0.38   | –44%   | <b>+11.3%</b> | 1.88 (.06)  | –0.08 |
| High-conviction long | +30.4% | 0.68   | –67%   | +31.9%        | 2.19 (.03)  | 0.96  |
| All buys long        | +22.6% | 0.67   | –73%   | +15.2%        | 1.88        | 0.98  |
| XBI                  | +11.9% | 0.55   | –64%   | —             | —           | 1.00  |

- Event level: high conviction +4.0%/qtr vs XBI,  $p \approx .10$  (marginal)
- L/S has the best drawdown (–44% vs –64%) but trails XBI on return: it holds no market exposure
- Long books beat XBI mostly through full biotech beta
- Same ~11–13%/yr alpha as SIC; lower  $t$  from ~35% fewer events

walkforward\_gics\_biotech.py; gross of costs

# Conviction: full GICS universe (875 firms)

Same script, unchanged, rerun on the complete GICS universe (875 firms instead of 495)

| Event level (1 quarter vs XBI)    | <i>N</i> | Mean   | <i>p</i> |
|-----------------------------------|----------|--------|----------|
| All buys                          | 35,929   | +1.50% | .50      |
| Low conviction (score $\leq 1$ )  | 15,812   | +3.40% | .45      |
| High conviction (score $\geq 3$ ) | 6,218    | +0.04% | .99      |

| Portfolio (63-session hold) | Alpha/yr | <i>t</i> | Beta |
|-----------------------------|----------|----------|------|
| All buys                    | +10.2%   | 1.69     | 0.94 |
| High-conviction buys        | +14.8%   | 1.52     | 0.98 |
| L/S hi-conv – sells         | +7.2%    | 1.55     | 0.02 |

## Not confirmed on the full universe

The score no longer sorts returns (high conviction +0.04% vs low +3.40%), and L/S alpha falls to +7.2%/yr,  $t = 1.55$ : not significant.

# Insider transactions and company events

## Research question

Do the return patterns around these events carry over to trades entered later, after allowing for costs?

We studied returns around insider purchases and sales, financing filings and FDA approvals. We used XBI as the sector benchmark.

We assembled **445 companies** from **83 observed XBI holdings snapshots**, including former holdings. Stock return data run through December 2025.

We measure insider event returns from **transaction dates** and enter the trading simulations after **public filing dates**. Company filings use filing dates. For approvals, the recorded FDA action date is our proxy for publication.

# Five-session returns after transactions and filings

We start at the close on or before the transaction or filing date.  
Stock return minus XBI return, in percentage points, before costs.

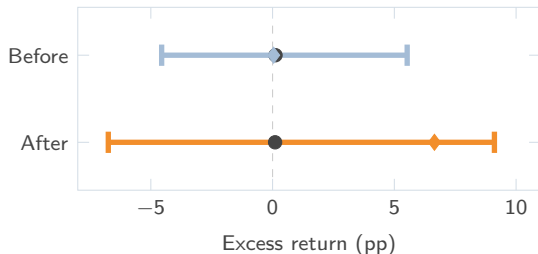
| Event             | <i>N</i> | Mean  | Median | Above XBI |
|-------------------|----------|-------|--------|-----------|
| Insider purchases | 8,767    | +3.64 | +1.67  | 59.8%     |
| Insider sales     | 33,264   | −0.33 | −0.53  | 45.7%     |
| S-1 registrations | 195      | −3.74 | −4.24  | 32.3%     |
| 424B5 supplements | 2,933    | −1.11 | −1.28  | 44.2%     |
| SC 14D-9 filings  | 89       | −0.06 | −0.05  | 49.4%     |

Purchases were followed by outperformance more often than sales. Both financing groups had negative average and median returns relative to XBI.

*N* counts company dates with complete five-session windows before and after, and both XBI and SPY returns available. Insider rows use nonderivative P/S transactions. Windows can overlap.

# Returns around FDA approvals

Approved original applications. Five sessions on each side, relative to XBI.



## After approval

Mean: **+6.65 pp**

Median: **+0.10 pp**

Above XBI: **50.8%**

187 observations  
52 issuers

Bars span the 10th to 90th percentiles. Circles mark medians.  
Diamonds mark means.

A few large gains lift the average. About half the observations beat XBI, and the median is close to zero.

The preceding window ends before the action date. The following window starts at the close on or before it.

# Entering one session later

We kept the same events and stocks, and moved both entry and exit one session later. Each window still spans five sessions.

| Event             | N     | Mean excess return vs XBI (pp) |              |
|-------------------|-------|--------------------------------|--------------|
|                   |       | Earlier window                 | Later window |
| Insider purchases | 2,348 | +2.48                          | +0.28        |
| S-1 registrations | 21    | −6.19                          | −1.66        |
| 424B5 supplements | 1,096 | −2.37                          | +0.27        |
| FDA approvals     | 165   | +1.50                          | +0.12        |

For purchases, the later window averaged 0.28 points above XBI. We use this later entry rule in the portfolio tests.

Earlier: close on or before the recorded date. Later: first close afterward. Insider dates are filing dates. FDA dates are action dates. We keep trades that meet the portfolio rules and have complete returns in both windows. Returns are before costs.

# How we formed the portfolios

**Eligibility and entry.** A company enters the universe when it first appears in an observed public XBI holdings filing, and stays eligible afterward. We enter trades at the first close after the signal date.

**Positions.** We target 50% stock exposure and 50% opposite XBI exposure, measured against portfolio equity. We weight the stocks equally and rebalance daily. We ignore new signals while a position is open.

**Costs.** We assume 10 basis points per leg on one-way turnover, plus 3% annual borrow on the short position. Cash earns zero.

**Holding periods.** We test 1, 2, 5, 10 and 20 sessions in both directions. Across nine event groups, this gives 90 portfolios. The groups also include merger proxies and public trial postings.

Each portfolio is a separate historical simulation. Insider signals use filing dates. FDA signals use action-date proxies. Trial signals use recorded public posting dates.

# Portfolio returns after costs

**58 of the 90 portfolios** had complete return histories. Of these, 22 gained before costs and **five after costs**. Those five appear below.

| Portfolio | Hold<br>(sessions) | Net return (%) |            | Max. drawdown<br>(%) | Trades |
|-----------|--------------------|----------------|------------|----------------------|--------|
|           |                    | Total          | Annualized |                      |        |
| S-1 short | 1                  | +3.78          | +0.19      | -12.04               | 21     |
| S-1 short | 2                  | +23.51         | +1.08      | -17.19               | 21     |
| S-1 short | 5                  | +6.90          | +0.34      | -25.93               | 21     |
| FDA long  | 1                  | +12.34         | +0.60      | -30.84               | 166    |
| FDA long  | 10                 | +20.15         | +0.94      | -41.90               | 164    |

May 2006–December 2025. Annualized returns include time in cash and use 252 trading sessions per year. The XBI position is opposite to the stock direction shown.

Annualized gains after costs ranged from 0.19% to 1.08%. Their largest drawdowns ranged from 12.04% to 41.90%. Each S-1 portfolio had 21 trades across 17 companies.

Missing returns during open positions leave 32 full-period results unresolved, including all insider purchase and sale portfolios.

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